

6(a)	$Q = It$	C1
	$= 0.80 \times 7.5 \times 60$	A1
	$= 360 \text{ C}$	
6(b)	$P = EI \text{ or } P = VI \text{ or } P = I^2R \text{ or } P = V^2 / R$	C1
	$0.80^2 \times 0.40 (= 0.256 \text{ W})$ or $0.48 \times 0.80 (= 0.384 \text{ W})$	C1
	efficiency = $(0.256 / 0.384) \times 100$ $= 67\%$	A1
6(c)(i)	$n = 3.2 \times 10^{22} / (1.3 \times 10^{-7} \times 3.0)$ $= 8.2 \times 10^{28} \text{ m}^{-3}$	A1
6(c)(ii)	$I = Anvq$	C1
	$v = 0.80 / (1.3 \times 10^{-7} \times 8.2 \times 10^{28} \times 1.60 \times 10^{-19})$	
	$= 4.7 \times 10^{-4} \text{ m s}^{-1}$	A1
6(d)	(wire Y has) larger resistance / resistance increases	M1
	(wire Y has) smaller current / current decreases	M1
	(average drift) speed is less (in wire Y)	A1